CS 4375 HW2

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1 Theoretical Part

1.1 Gradient Descent

Single example, target t, learning rate η, squared error:

$$o = w_0 + \sum_{j=1}^{n} w_j (x_j + x_j^2),$$
$$E = \tfrac{1}{2}(t - o)^2.$$

By the chain rule,

$$\frac{\partial E}{\partial w_i} = (t - o)\,\frac{\partial (t - o)}{\partial w_i}$$
$$= -(t - o)\,\frac{\partial o}{\partial w_i}.$$

Only the i-th term of o depends on w_i, and w_0 appears alone:

$$\frac{\partial o}{\partial w_0} = 1,$$
$$\frac{\partial o}{\partial w_i} = x_i + x_i^2.$$

Hence

$$\frac{\partial E}{\partial w_0} = -(t - o),$$
$$\frac{\partial E}{\partial w_i} = -(t - o)(x_i + x_i^2).$$

Using $\Delta w_i = -\eta\, \partial E / \partial w_i$:

$$w_0 \leftarrow w_0 + \eta\,(t - o),$$
$$w_i \leftarrow w_i + \eta\,(t - o)(x_i + x_i^2), \qquad i = 1, \ldots, n.$$

Over a training set D,

$$E = \tfrac{1}{2} \sum_{d \in D} (t_d - o_d)^2$$

and each update is summed over $d \in D$.

1.2 Comparing Activation Functions

Assumptions: no biases in (a), (b); $w_{j,i}$ connects neuron i to neuron j; the input layer is the identity.

(a) Hidden layer:

$$\begin{aligned} a_3 &= w_{3,1}x_1 + w_{3,2}x_2, & y_3 &= h(a_3),\\ a_4 &= w_{4,1}x_1 + w_{4,2}x_2, & y_4 &= h(a_4). \end{aligned}$$

Output:

$$\begin{aligned} y_5 &= h\big(w_{5,3}\,y_3 + w_{5,4}\,y_4\big)\\ &= h\Big(w_{5,3}\,h(w_{3,1}x_1 + w_{3,2}x_2) + w_{5,4}\,h(w_{4,1}x_1 + w_{4,2}x_2)\Big). \end{aligned}$$

(b) With

$$\begin{aligned} X &= \begin{pmatrix} x_1 \\ x_2 \end{pmatrix},\\ W^{(1)} &= \begin{pmatrix} w_{3,1} & w_{3,2} \\ w_{4,1} & w_{4,2} \end{pmatrix},\\ W^{(2)} &= \begin{pmatrix} w_{5,3} & w_{5,4} \end{pmatrix}, \end{aligned}$$

and h applied elementwise,

$$\begin{aligned} W^{(1)}X &= \begin{pmatrix} a_3 \\ a_4 \end{pmatrix},\\ h(W^{(1)}X) &= \begin{pmatrix} y_3 \\ y_4 \end{pmatrix},\\ W^{(2)}h(W^{(1)}X) &= w_{5,3}y_3 + w_{5,4}y_4, \end{aligned}$$

so

$$y_5 = h\big(W^{(2)}\,h(W^{(1)}X)\big).$$

(c) Assumption: biases $w_{j,0}$ are allowed (constant input 1). In Figure 1 the hidden units are $j \in \{3,4\}$ and the output unit is 5. Write

$$\begin{aligned} a_j &= w_{j,1}x_1 + w_{j,2}x_2 + w_{j,0} \qquad (j = 3,4),\\ a_5 &= w_{5,3}\,h(a_3) + w_{5,4}\,h(a_4) + w_{5,0},\\ y_5 &= h(a_5). \end{aligned}$$

Relation between h_s and h_t. Multiplying top and bottom of h_t by e^{-x}:

$$\begin{aligned} h_t(x) &= \frac{e^x - e^{-x}}{e^x + e^{-x}} \\ &= \frac{1 - e^{-2x}}{1 + e^{-2x}} \\ &= \frac{2 - (1 + e^{-2x})}{1 + e^{-2x}} \\ &= \frac{2}{1 + e^{-2x}} - 1 \\ &= 2h_s(2x) - 1, \end{aligned}$$

so

$$h_s(x) = \tfrac{1}{2}\, h_t\big(\tfrac{x}{2}\big) + \tfrac{1}{2}.$$

I. *Sigmoid network $\Rightarrow$ tanh network.* Let $h = h_s$ with weights w. Then $h_s(a_j) = \frac{1}{2}h_t(a_j/2) + \frac{1}{2}$ and

$$\begin{aligned} a_5 &= \sum_{j=3}^{4} w_{5,j}\, h_s(a_j) + w_{5,0} \\ &= \sum_{j=3}^{4} w_{5,j}\Big[\tfrac{1}{2}h_t\big(\tfrac{a_j}{2}\big) + \tfrac{1}{2}\Big] + w_{5,0} \\ &= \sum_{j=3}^{4} \tfrac{w_{5,j}}{2}\, h_t\big(\tfrac{a_j}{2}\big) + w_{5,0} + \tfrac{1}{2}\sum_{j=3}^{4} w_{5,j}, \\ y_5^{(s)} &= h_s(a_5) \\ &= \tfrac{1}{2}\, h_t\big(\tfrac{a_5}{2}\big) + \tfrac{1}{2}. \end{aligned}$$

Now take a tanh network ($h = h_t$) with weights u:

$$\begin{aligned} u_{j,i} &= \tfrac{1}{2}w_{j,i} \qquad (j = 3, 4;\ i = 1, 2), \\ u_{j,0} &= \tfrac{1}{2}w_{j,0} \qquad (j = 3, 4), \\ u_{5,j} &= \tfrac{1}{4}w_{5,j}, \\ u_{5,0} &= \tfrac{1}{2}\Big(w_{5,0} + \tfrac{1}{2}\textstyle\sum_{j=3}^{4} w_{5,j}\Big). \end{aligned}$$

Its hidden inputs are

$$b_j = \sum_i u_{j,i}x_i + u_{j,0} = \tfrac{a_j}{2},$$

so its hidden outputs are $h_t(a_j/2)$. Then

$$\begin{aligned}
b_5 &= \sum_{j=3}^{4} u_{5,j}\, h_t\big(\tfrac{a_j}{2}\big) + u_{5,0} \\
&= \sum_{j=3}^{4} \tfrac{w_{5,j}}{4}\, h_t\big(\tfrac{a_j}{2}\big) + \tfrac{w_{5,0}}{2} + \tfrac{1}{4}\sum_{j=3}^{4} w_{5,j} \\
&= \tfrac{1}{2}\Big[\sum_{j=3}^{4} \tfrac{w_{5,j}}{2}\, h_t\big(\tfrac{a_j}{2}\big) + w_{5,0} + \tfrac{1}{2}\sum_{j=3}^{4} w_{5,j}\Big] \\
&= \tfrac{a_5}{2}, \\
y_5^{(t)} &= h_t(b_5) \\
&= h_t\big(\tfrac{a_5}{2}\big)\,,
\end{aligned}$$

hence

$$y_5^{(s)} = \tfrac{1}{2}\, y_5^{(t)} + \tfrac{1}{2}.$$

II. *Tanh network ⇒ sigmoid network.* Start from a tanh network with weights u, hidden inputs b_j and output input b_5. Using $h_t(b_j) = 2h_s(2b_j) - 1$, define $z_j = h_s(2b_j)$ for $j = 3, 4$, so $h_t(b_j) = 2z_j - 1$ and

$$\begin{aligned}
b_5 &= \sum_{j=3}^{4} u_{5,j}\,(2z_j - 1) + u_{5,0} \\
&= \sum_{j=3}^{4} 2u_{5,j}\, z_j + u_{5,0} - \sum_{j=3}^{4} u_{5,j}.
\end{aligned}$$

Take a sigmoid network with weights

$$\begin{aligned}
w_{j,i} &= 2u_{j,i} \qquad (j = 3, 4;\ i = 1, 2), \\
w_{j,0} &= 2u_{j,0} \qquad (j = 3, 4), \\
w_{5,j} &= 4u_{5,j}, \\
w_{5,0} &= 2u_{5,0} - 2\textstyle\sum_{j=3}^{4} u_{5,j}.
\end{aligned}$$

Its hidden inputs are $a_j = 2b_j$, so its hidden outputs are z_j, and its output input is

$$\begin{aligned}
a_5 &= \sum_{j=3}^{4} 4u_{5,j}\, z_j + 2u_{5,0} - 2\sum_{j=3}^{4} u_{5,j} \\
&= 2b_5.
\end{aligned}$$

Therefore $y_5^{(s)} = h_s(2b_5)$, and since $h_t(b_5) = 2h_s(2b_5) - 1$,

$$y_5^{(t)} = 2\, y_5^{(s)} - 1.$$

(Note that this weight map is the inverse of the one above.)

In conclusion, every sigmoid network of this shape has a tanh network with

$$y^{(s)} = \tfrac{1}{2} y^{(t)} + \tfrac{1}{2},$$

and every tanh network has a sigmoid network with

$$y^{(t)} = 2y^{(s)} - 1.$$

The weights are related by scalings and a bias shift, so the two families compute the same functions up to an affine map of the output.